

CAIE Course Program

Assignment 4 - Basic FastAPI App | learn. build. operate

1. What you will build

Build a **very basic web API** using the **FastAPI** framework. FastAPI lets you create web endpoints in Python that return data (usually JSON) when someone visits a URL. Your task is to set up a clean Python environment, install FastAPI, write a small app in a **.py** file, run it with a server, and view it working in the browser. FastAPI also generates interactive API documentation for you automatically.

Difficulty: Beginner · Tools: Python, FastAPI, Uvicorn

2. What your app must include

- Use the **FastAPI** framework to build the app.
- Create a root endpoint GET `/` that returns a simple JSON response (for example, a welcome message).
- Add at least one more endpoint that takes a value from the URL - either a path parameter (for example, `/greet/{name}`) or a query parameter - and returns it inside the response.
- Run the app with the Uvicorn server and confirm it works in the browser.
- Open the automatic interactive docs at `/docs` and check your endpoints there.
- Keep all your code in a single file named `main.py`.

Think it through

FastAPI builds the interactive docs from your code, so clear endpoint names and return values matter. Think about what each endpoint should return, what happens when a parameter is missing, and how the JSON response should be shaped. Keep it simple - this is meant to be a first, small API.

3. Step-by-step setup

Follow these steps in order. Commands are shown for both macOS/Linux and Windows where they differ.

Step 1 - Create a project folder

```
mkdir fastapi-basic-app
cd fastapi-basic-app
```

Step 2 - Create a virtual environment (venv)

A virtual environment keeps this project isolated from the rest of your system.

```
python -m venv venv  
# (on some systems use 'python3' instead of 'python')
```

Step 3 - Activate the virtual environment

```
# macOS / Linux:  
source venv/bin/activate  
  
# Windows (PowerShell):  
venv\Scripts\Activate.ps1  
  
# Windows (Command Prompt):  
venv\Scripts\activate.bat
```

When activated, you will see (venv) at the start of your terminal prompt.

Step 4 - Install FastAPI and the server

Install FastAPI together with Uvicorn, the server that runs your app.

```
pip install fastapi uvicorn
```

Step 5 - Run your app

Start the app with Uvicorn. Replace `file_name` with the name of your Python file and app with your FastAPI variable name.

```
uvicorn file_name:app --reload
```

The app runs at `http://127.0.0.1:8000`, and the interactive docs are at `http://127.0.0.1:8000/docs`.

Step 6 - Stop and deactivate when finished

Press `Ctrl + C` in the terminal to stop the server, then deactivate the venv.

```
deactivate
```

4. How it should behave

Your wording can differ, but visiting the root endpoint should return JSON similar to this:

```
GET http://127.0.0.1:8000/  
  
{  
  "message": "Hello, FastAPI"  
}
```

5. What to submit

- Push your code to **GitHub** (a single repository for this app).
- Share your GitHub repository link in **WhatsApp**.
- Post your screen-recorded video of the app working in the **Skool community**.

Reminder: build the app yourself - the steps above are your setup roadmap, not the answer. Good luck!